

# AMAR DHILLON

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## Education

### University of Maryland, College Park

*B.S. in Computer Science, Minor in Statistics; GPA: 3.5*

Expected May 2028

*College Park, MD*

## Projects

### IEX Order Book & Matching Engine | *C++17, STL, POSIX, Linux perf*

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Sustained **1.26M events/sec** over the full replay and a 1.2  $\mu$ s per-message p99 on a separate 2M-event run, keeping each price level as a FIFO queue (`std::list`) inside a `std::map`, with an order-ID hash map for  $O(1)$  cancels
- Implemented B+ tree and LSM-tree backends for the event log and benchmarked them on identical 2M-record workloads, where the B+ tree gave **12.8x** the read throughput and 22x faster range queries at the cost of 27% more disk space
- Traced an apparent 1.65x regression to the CPU frequency governor, not a code change, using perf counters and a do-nothing control book, then cut run-to-run variance from about 6% to 0.06% by pinning cores and fixing the clock

## Experience

### Amazon Web Services

Sep 2026 – Dec 2026

*Software Development Engineer Intern*

*Redmond, WA*

- Building capacity planning for Amazon Redshift maintenance like patching: Java logic that sets upper and lower bounds on reserved EC2 capacity to balance availability against cost, validated by backtesting

### Capital One

Jun 2026 – Aug 2026

*Software Engineering Intern*

*Richmond, VA*

- Cut AI workflow generation time **65%** to under 15s by replacing a Claude skill's open-ended search with focused sub-skills that call Python tools for workflow search and validation, guided by CloudWatch and OpenSearch latency logs
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a category tree that the Claude skill searches first, with Redis caching full workflows, so lookups rarely touch a database gated by permissions and rate limits

### UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

*Software Engineer (Contract)*

*College Park, MD*

- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda

### UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

*Machine Learning Research Intern*

*College Park, MD*

- For a PhD-led robotics project aimed at ICLR, built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D for a vision-language model, then tuned prompts so it favors grasps that avoid hitting the table

### TerpLabs

Jun 2025 – Present

*Co-Founder, Software Engineer*

*College Park, MD*

- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency

## Technical Skills

**Languages:** C++17, C, Python, Java, Rust, SQL, TypeScript, Bash

**Systems & Performance:** debugging and profiling with Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees, data structures and algorithms, benchmarking

**Cloud & Tools:** AWS (Lambda, Kinesis, EC2, Fargate, CloudWatch monitoring), AWS CDK, PostgreSQL, Redis, Docker, GitHub Actions (CI/CD), Git, Pytest

**Certification:** AWS Certified Developer – Associate (Nov 2024 – Nov 2027)